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New Virtual and Paleomagnetic Pole Positions from Isotopically Dated Precambrian Rocks in Wyoming, Montana, and Arizona: Their Significance in Establishing a North American Apparent Polar Wandering Path

ABSTRACT

Paleomagnetic directions have been determined from well-dated Precambrian rocks, in large part mafic dikes, in Wyoming, Montana, and Arizona. Ages, determined radiometrically, range from 700 to 2,550 m.y. When these data, together with the compilation of acceptable well-dated paleomagnetic poles from North America, are treated by a running-mean method, a generalized belt of apparent polar wandering is suggested. This belt is simple in form, and the rate of polar wandering is nearly uniform. The existence of large-scale loops of relatively rapid wander velocity is largely unsupported.

INTRODUCTION

Potentially the most quantitative and useful technique for reconstructing large-scale tectonic events such as continental drift and plate movement in the Precambrian involves matching of apparent polar wandering paths from two or more land masses. Unfortunately, because of the paucity of well-dated paleomagnetic poles from the various continents, this can be done only rather poorly at present. As part of the solution to this difficulty, an apparent polar wandering path should first be unambiguously defined for one continent. As portions of a wander path from other continents are obtained, better attempts at matching can be made.

In view of the abundance of radiometrically dated igneous rocks from North America, it appears to be the best place to delineate a well-defined polar wandering path for much of Precambrian time. There have been several attempts toward this end (Dubois, 1962; Black, 1963; Strangway, 1964; Spall, 1971a; Robertson and Fahrig, 1971; Irving and others, 1972c; Irving and Park, 1972). From the most recent compilations, it is apparent that certain portions of the path are relatively well known and generally acceptable. The points of most controversy center on the reality and location of large-scale loops of relatively rapid polar wandering.

The apparent polar wandering path of Spall (1971a) shows two loops: one covering the time span of 2,500 to 1,500 m.y., and one from 1,200 to 1,000 m.y. (Fig. 1). In contrast to this curve is the Great Logan Loop, covering a time interval from about 1,200 to 700 m.y. (Robertson and Fahrig, 1971; Fig. 1). The path given by Irving and others (1972c) includes the Great Logan Loop back to 1,200 m.y. and a loop from 2,700 to 1,500 m.y. ago (Fig. 1). These loops, if real, would be extremely important geologically and geophysically. Various authors have associated periods of rapid wander (periods of rapid or changing continental movement) with tectonic, igneous, and metamorphic events (Gastil, 1960; Fahrig and Wanless, 1963; Sutton, 1963; Engel, 1971; Muehlberger, 1971; and Irving and Park,

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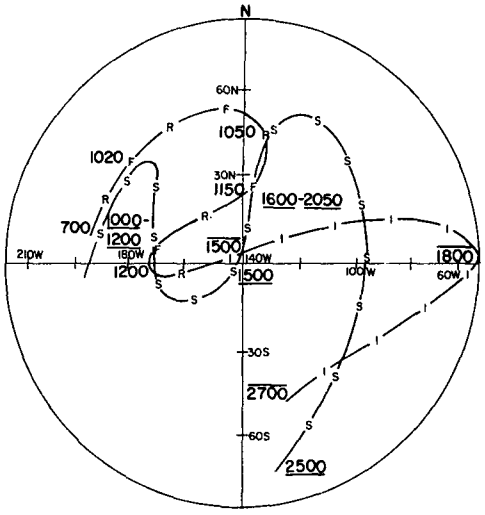


Figure 1. Apparent paleomagnetic polar wander curves from Spall (1972) = S; Robertson and Fahrig (1971) = R-F; and Irving and others (1972c) = I; equal-area projection. Numbers are in millions of years: underlined, Spall; overlined, Irving; unlined, Robertson and Fahrig.

1972) and with growth periods in the Earth's core (Runcorn, 1962). Furthermore, reconstruction of crustal movements has been based on the location of these loops (Irving and others, 1972a). If, on the other hand, these large loops do not actually exist, and the rate of polar wandering is relatively uniform, this too would be significant.

NEWLY OBTAINED RESULTS FROM NORTH AMERICA

General

New Precambrian data have been obtained from twenty isotopically dated mafic dikes in the Beartooth Mountains, Wyoming and Montana, the Bighorn Mountains, and Owl Creek Mountains, Wyoming (Fig. 2), from four diabase intrusive masses and unbaked Hakatai Shale near Hance Rapids in the Grand Canyon National Park, Arizona, and from an anorthosite intrusive mass in the Laramie Range near Laramie, Wyoming. Ages of these rocks range from about 700 m.y. to at least 2,500 m.y. and perhaps as much as 2,800 to 3,000 m.y. About two-thirds of the intrusive rocks collected produced fair to excellent paleomagnetic results. Directions of magnetization were accepted as meaningful if they grouped relatively well ($k \geq 10$) and if both a.c. and thermal cleaning produced similar results (Table 1). Most dikes showed best grouping after a.c. treatments up to 300 Oe. Application of peak fields beyond that produced increasing scatter. Polished and thin sections were also studied for indications of secondary alteration. The results obtained from a few dikes were eliminated in accordance with these criteria.

Beartooth Mountains, Wyoming and Montana

The Beartooth Mountains are essentially an uplifted block of Precambrian crystalline rocks intruded by Precambrian mafic dikes and Tertiary stocks (Harris, 1959; Foose and others, 1961; Prinz, 1964, 1965; Casella, 1964, 1969). Precambrian deformation is discussed by Harris (1959); Casella (1964, 1969); Rowan and Larsen (1970); Rowan and Mueller (1971); and McMannis and others (1971a, 1971b). None of the dikes where sampled shows evidence of extensive folding (Rowan and Mueller, 1971). Isolated remnants of gently dipping early Paleozoic sedimentary rocks occur on top of the Beartooth Precambrian plateau.

All eight mafic dikes sampled (Fig. 3) have been radiometrically dated either by P. Mueller (1972, personal commun.) by whole-rock K-Ar and (or) whole-rock Rb-Sr isochron methods; or by Condie and others (1969) by whole-rock K-Ar analysis; or both. Table 2 is a summary of the dikes from which we collected samples, their ages, and source of information.

Dikes BT-7 and BT-8 have been dated both by whole-rock K-Ar and Rb-Sr (whole-rock isochron) methods as about 1,500 m.y. (P.

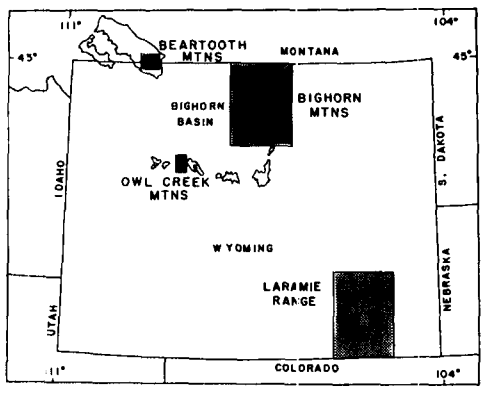


Figure 2. Regional index map of Wyoming and Montana showing sampling areas.

TABLE 1. SUMMARY OF PALEOMAGNETIC DATA

Unit	Treat	N	$\bar{I}$	$\bar{D}$	K	α_{95}	Lat	Long	δp	δm
BEARTOOTH MOUNTAINS										
BT-1	NRM	6	55	328	45	10	63 N.	208	10	14
	100oe	7	71	19	47	9	74 N.	69	13	15
	200oe	7	70	337	14	17	73 N.	160	24	28
	400°C	4	-46	347						
	500°C	7	77	189						
BT-2a	NRM	9	5	284	8	19	12 N.	207	10	19
	50oe	3	4	287	8	47	13 N.	208	23	47
	100oe	6	- 8	289	13	19	11 N.	216	10	19
	200oe	3	-12	293	17	30	11 N.	220	15	31
	400°C	6	49	94	7	27	18 N.	44	23	35
	500°C	6	62	82	7	27	34 N.	49	33	42
BT-2b	NRM	5	55	262	4	43	19 N.	168	43	61
	100oe	5	6	288	14	22	14 N.	210	11	22
	200oe	5	-11	280	14	26	3 N.	210	13	26
	400°C	4	- 1	277	6	40	5 N.	204	20	40
	500°C	5	- 3	316	3	59	15 N.	246	39	69
BT-3	NRM	6	44	139	6	31	10 S.	72	24	39
	100oe	5	44	129	49	11	6 S.	65	9	14
	200oe	5	40	130	63	9	9 S.	64	7	12
	400°C	5	34	132	108	7	13 S.	64	5	9
	500°C	5	42	126	60	10	6 S.	62	7	12
BT-4	NRM	6	55	76	6	31	33 N.	38	31	44
	100oe	6	26	62	5	32	29 N.	9	19	34
	200oe	6	20	107	18	16	5 S.	39	9	17
	300°C	6	24	112	16	17	6 S.	44	10	18
	400°C	6	18	110	51	10	7 S.	40	5	10
	500°C	6	16	87	7	27	8 N.	23	14	28
BT-5	NRM	9	39	113	4	31	1 N.	51	22	37
	100oe	9	18	117	22	11	12 S.	45	6	12
	200oe	9	16	112	12	15	10 S.	41	8	16
	400°C	7	10	121	10	21	18 S.	46	11	21
	500°C	8	14	117	55	8	14 S.	44	4	8
BT-6	NRM	7	14	132	3	40	23 S.	57	21	41
	100oe	7	-26	122	23	13	32 S.	33	7	14
	200oe	6	-35	129	40	11	40 S.	34	7	12
	400°C	6	-34	124	8	25	37 S.	30	16	28
	500°C	6	-44	121	17	16	39 S.	21	13	20
BT-7	NRM	6	-56	297	6	31	9 N.	62	32	45
	100oe	6	-61	326	35	11	2 N.	85	13	18
	200oe	6	-59	338	84	7	3 S.	92	8	11
	400°C	6	-61	340	73	8	1 S.	95	9	12
	500°C	5	-65	340	15	21	3 N.	96	27	33
BT-8	NRM	6	- 8	25	2	83	37 S.	141	42	83
	100oe	6	-61	346	51	10	2 S.	99	15	25
	200oe	6	-49	2	24	14	16 S.	111	12	19
	400°C	6	-55	1	22	15	10 S.	111	15	21
	500°C	5	-47	4	18	19	17 S.	113	15	24
BT-9	NRM	5	68	351	32	14	81 N.	149	19	23
	100oe	5	65	340	289	5	76 N.	183	6	7
	200oe	5	61	343	206	5	77 N.	205	6	8
NORTHERN BIGHORN MOUNTAINS										
NBH-1	NRM	8	40	332	1					
	100oe	7	53	76	2	84	32 N.	35		
	200oe	5	39	106	3	49	5 N.	44	35	59
	300oe	3	39	127	5	60	7 S.	59	42	71
	400°C	7	48	343	4	34	70 S.	62	29	45
	500°C	8	39	332	9	20	56 S.	55	13	23
NBH-2	NRM	8	-63	134	4	34	57 S.	2		
	100oe	2	-41	149	1		57 S.	48		
	200oe	6	-80	146	1		60 N.	130		
	400°C	4	-51	186	7	39	76 S.	127	35	53
	500°C	4	-50	184	6	42	76 S.	121	37	56
NBH-3	NRM	3	-17	316	24	26	23 S.	59	14	26
	100oe	7	- 1	259	2	83	8 N.	10	41	83
	200oe	5	-33	213	1		52 S.	164		
	400°C	6	44	141	1		11 S.	72		
NBH-4	NRM	5	46	153	5	41	14 S.	83	43	52
	100oe	6	0	148	14	19	37 S.	65	9	19
	400°C	6	-34	140	7	28	48 S.	42	18	32
	500°C	4	-47	123	6	43	43 S.	18	36	56
NBH-5	NRM	6	71	339	29	13	73 N.	153	19	22
	400°C	8	69	344	31	10	77 N.	159	14	17
	500°C	8	65	342	44	8	77 N.	179	11	14
NBH-6	NRM	5	15	194	51	11	37 S.	125	6	11
	100oe	5	- 5	187	9	27	47 S.	118	14	27
	200oe	6	-27	183	24	14	59 S.	113	8	15
	400°C	6	10	172	3	47	40 S.	97	24	47
	500°C	6	58	120	2	62	9 N.	64	67	91
SOUTHERN BIGHORN MOUNTAINS										
SBH-1	NRM	5	-19	221	14	22	41 S.	165	12	23
	100oe	6	-44	206	15	18	62 S.	164	14	23
	400°C	7	-49	211	17	15	61 S.	174	12	19
	500°C	7	-48	201	21	14	67 S.	164	12	18
SBH-2	NRM	7	3	159	2	69	40 S.	79	35	70
	100oe	7	-22	170	2	82	56 S.	90	46	87
	200oe	5	-43	172	16	20	70 S.	86	15	25
	400°C	6	-26	178	1					
	500°C	5	-51	178	22	17	78 S.	100	15	23
SBH-3	NRM	6	34	219	6	31	18 S.	146	20	35
	100oe	3	11	201	5	67	37 S.	133	34	68
	400°C	8	24	202	5	28	31 S.	132	16	29
	500°C	8	6	191	5	28	42 S.	122	14	28
OWL CREEK MOUNTAINS										
OC-1	NRM	3	- 4	200	4	67	45 N.	317	34	
	100oe	7	-46	29	1					
	200oe	7	-45	27	1					
	400°C	3	-64	6	1					
OC-2	NRM	6	-74	16	2	74	14 N.	117		
	100oe	6	-37	12	10	22	25 S.	120	15	36
	200oe	4	-33	6	11	29	29 S.	115	19	33
	300oe	2	-34	1	6					
	400°C	6	-32	355	18	16	29 S.	102	10	18
	500°C	5	-40	5	12	23	24 S.	114	17	28
OC-3	NRM	6	57	26	1					
	100oe	6	-44	351	1				83	
	200oe	6	-30	20	2	94	28 S.	131	57	
	300oe	3	-26	66	1		7 S.	172		
	400°C	5	-31	11	16	20	29 S.	120	12	22
	500°C	2	-77	46	2		24 N.	128		
OC-4	NRM	8	-29	335	3	36	29 S.	91	22	40
	100oe	6	-40	6	150	5	24 S.	114	4	7
	400°C	8	-42	4	92	6	22 S.	112	4	7
	500°C	6	-41	3	626	3	23 S.	112	2	3
	550°C	2	-38	12	167	19	24 S.	121	13	23
LARAMIE RANGE										
IM-1	NRM	8	-38	51	20	13	12 S.	333	9	15
	100oe	8	-34	50	110	5	14 S.	334	3	6
	400°C	8	-36	49	64	4	14 S.	332	5	8
	500°C	8	-35	50	70	4	13 S.	334	4	8
GRAND CANYON, ARIZONA										
HR-1	NRM	7	35	268	261	4	10 N.	185	3	4
	50oe	7	26	264	136	5	3 N.	187	3	6
	100oe	7	20	262	62	8	1 S.	189	8	25
	400°C	7	19	260	45	9	3 S.	188	5	10
	500°C	7	22	259	126	5	2 S.	186	3	6
HR-4	NRM	9	35	263	30	10	6 N.	182	6	11
	50oe	9	30	257	41	8	1 S.	182	5	9
	100oe	9	28	255	42	8	3 S.	181	5	9
	400°C	9	23	255	44	8	5 S.	183	4	8
	550°C	9	22	255	4	28	5 S.	184	16	30
HR-5	NRM	9	36	263	23	11	6 N.	182	8	13
	50oe	9	35	263	24	11	5 N.	182	7	12
	100oe	9	32	261	26	10	3 N.	183	7	12
	500°C	7	22	262	18	15	0	188	8	15
HR-6	NRM	7	32	269	158	5	9 N.	187	3	5
	50oe	7	26	268	137	5	6 N.	190	3	6
	100oe	7	23	268	134	3	5 N.	191	3	6
	400°C	7	22	267	150	5	4 N.	191	3	5
	500°C	7	10	269	31	11	2 N.	197	6	11
HR-3	NRM	6	52	258	34	12	10 N.	169	11	16
	100oe	6	49	257	45	10	8 N.	170	9	13
	500°C	2	18	250	10	92	11 S.	182	49	96

Note: Treat refers to demagnetization values and technique: AF demagnetization in oersteds and temperature of thermal demagnetization in degrees Centigrade. N is the number of specimens. $\bar{I}$ is the mean inclination positive downward. $\bar{D}$ is the mean declination of the north seeking magnetic vector, east of true north. K is Fisher's (1953) best estimate of dispersion. α_{95} is the semi-angle of the cone of 95 percent level confidence. Lat is the latitude of the paleomagnetic pole position. Long is the longitude in degrees west of the paleomagnetic pole position. δp and δm are the semiminor and semimajor axes of the oval of 95 percent level confidence.

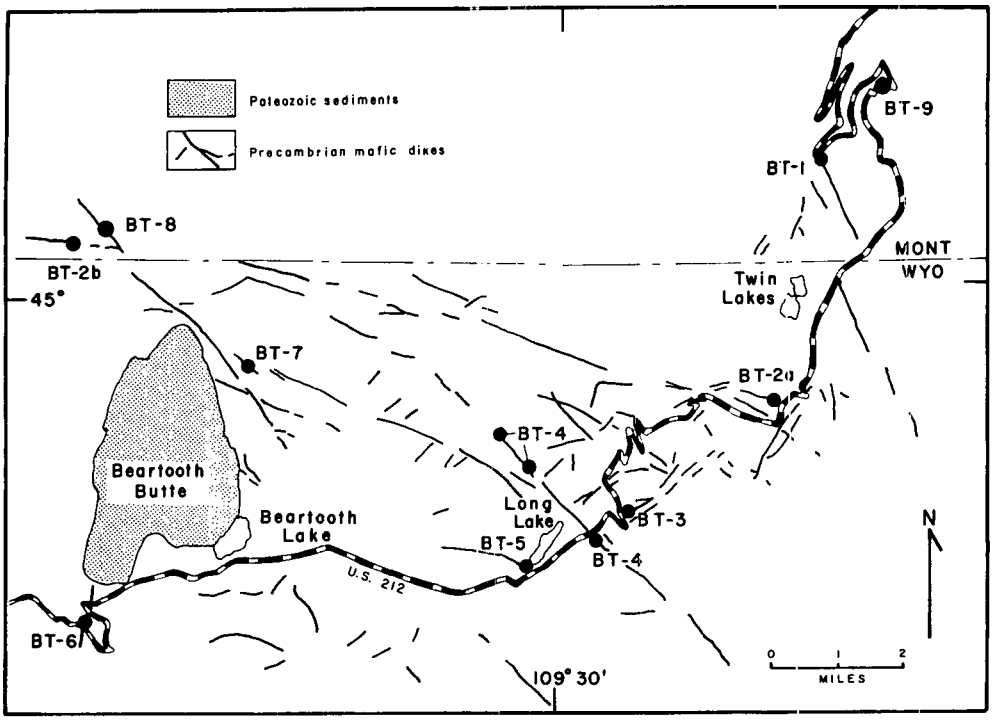


Figure 3. Location map of Beartooth Mountains, Wyoming and Montana sampling sites (after Prinz, 1964).

Mueller, 1972, personal commun.). They are unrecrystallized and give consistent directions at all demagnetization values (Table 1; Fig. 4). The virtual paleomagnetic pole position for these dikes is 4° S., 96° W., a position about 50° east of the accepted 1,500 m.y. paleomagnetic pole position (Fig. 5). Although secular variation could account for some of the angular separation, a significant measure of structural rotation appears to be additionally necessary. We have no means of knowing the amount of secular dispersion, and therefore we have assumed that all 50° is the result of simple rotation. Since the Cambrian sediments of nearby Beartooth Butte are essentially flat lying, the movement must have occurred during the Precambrian. If we rotate westward the pole for these dikes, we must do so for all older dikes in this general region. There is no independent structural evidence in support of or in opposition to this interpretation. The validity of this hypothesis will be tested as more paleomagnetic pole positions from well-dated rocks 1,600 to 2,600 m.y. become available.

Dike BT-6, dated by Rb-Sr as 2,550 m.y. old and by K-Ar as 2,000 m.y. old, is the only 2,550 m.y. unrecrystallized dike collected. Its paleomagnetic direction is somewhat different from the other 2,550-m.y.-old dikes, all of which have been recrystallized (Fig. 4). For this reason, we conclude that the paleomagnetic direction of dike BT-6 corresponds to the field direction 2,550 m.y. ago. The corresponding virtual geomagnetic pole position is 40° S., 27° W. (Table 1); rotation of this pole westward by 50° brings the virtual geomagnetic

TABLE 2. RADIOMETRIC AGES OF DIKES FROM BEARTOOTH MOUNTAINS

Dike	Whole-rock* (age in m.y.)	Whole-rock (age in m.y.)
BT-1	2,550 Rb-Sr	1,010 K-Ar
BT-2	730 K-Ar	730 K-Ar
BT-3	2,550 Rb-Sr	2,160 K-Ar
BT-4	2,550 Rb-Sr	1,690 K-Ar
BT-5	2,550 Rb-Sr	1,960 K-Ar
BT-6	2,550 Rb-Sr	2,000 K-Ar
BT-7	1,500 Rb-Sr, K-Ar	1,500 K-Ar
BT-8	1,500 Rb-Sr, K-Ar	

* Dated by Mueller, 1972, personal commun.
† Dated by Condie and others, 1969.

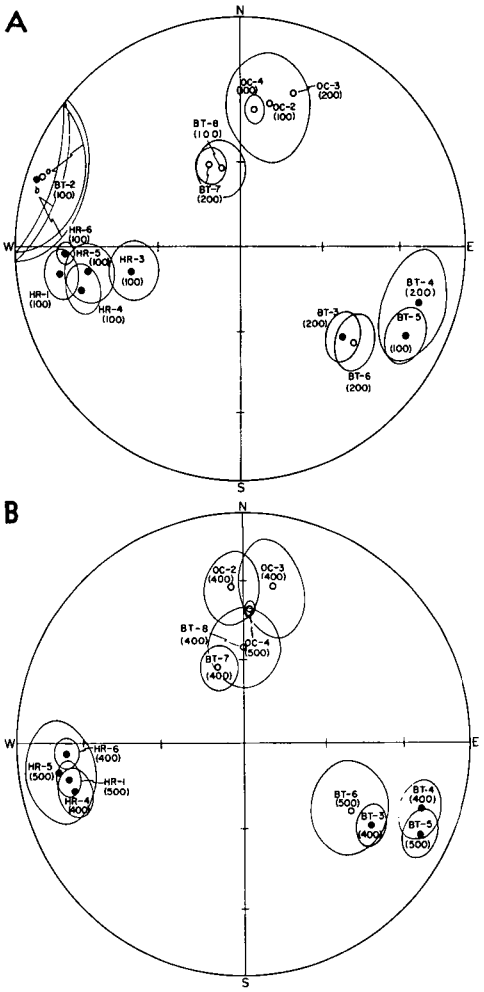


Figure 4. Mean directions of magnetization and 95 percent ovals of confidence for dikes from the Beartooth Mountains (BT), Owl Creek Mountains (OC), and Hance Rapids area (HR); equal-area projection; solid circles denote downward, open circles denote upward. A. Mean directions after alternating current demagnetization. Peak fields in oersteds used to produce maximum K values given in parentheses. B. Mean directions after thermal demagnetization. Temperatures at which K values become maximum are given in parentheses.

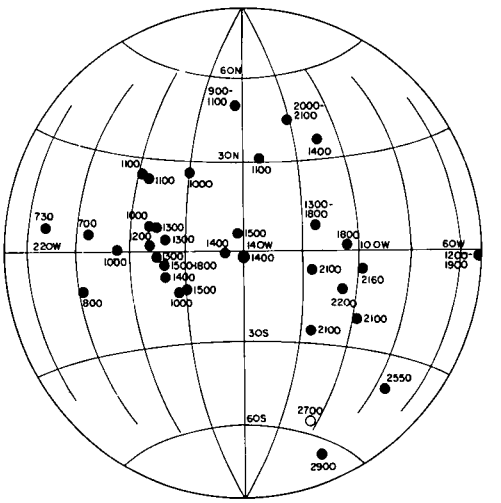


Figure 5. Virtual and paleomagnetic pole positions from literature and our study; equal-area projection. Solid dot, front hemisphere; circle, back hemisphere. Ages given in millions of years. References for pole positions are: <1.0 b.y.—Robertson and Baragar (1972), Fahrig and others (1971), Park and Irving (1972); ~1.0 to 1.1 b.y.—Murthy and Deutsch (1972), Faure (1964), Spall (1971a), Beck and Lindsley (1969), Fahrig and Jones (1969), Jahrein (1965), Vincenz (1968), Palmer (1970), Robertson and Fahrig (1971), Spall (1970), DuBois (1962); 1.2 to 1.6 b.y.—Eggler and Larson (1968), Helsley and Spall (1971), Fahrig and others (1965), Hsu and others (1966), Hays and Scharon (1966), Palmer (1969), Spall (1968), Kellogg (1972), Murthy and others (1965), Fahrig and Jones (1969), Irving and others (1972b, 1972c), Larochelle (1969); >1.6 b.y.—Spall (1971b), Symons (1967, 1970), Murthy and Deutsch (1972), Bergh (1970), Gates (1972), Strangway (1964).

pole to 40° S., 77° W., which is quite close to pole positions of the 2,600- to 2,900-m.y.-old Matachewan and Stillwater intrusives (Fig. 5).

Dikes BT-3, BT-4, and BT-5 are all 2,550 m.y. old, according to a whole-rock Rb-Sr isochron and between 1,690 and 2,160 m.y. by the whole-rock K-Ar method; all (as noted above) have been recrystallized. Directions of magnetization of all three are well grouped (Fig. 4), consistent at all demagnetization levels, and yield similar virtual paleomagnetic pole positions (Table 2). It is probable that the remanence directions represent the geomagnetic field at the time of recrystallization. This must be younger than the Rb-Sr age of 2,550 m.y. and probably close to the maximum K-Ar age of 2,160 m.y. Rotation of the average paleomagnetic pole position westward by 50° brings it to 5° S., 99° W., a direction that closely approximates that from dikes of equivalent age (Fig. 5).

Dike BT-1 is also 2,550 m.y. by Rb-Sr methods and is recrystallized; however, its K-Ar age is only 1,010 m.y. The sampling site is located a few meters from a porphyritic

acidic Tertiary stock which was sampled on a higher switchback of U.S. Highway 212 (BT-9). The directions of magnetization of the Precambrian mafic dike are closely similar to the directions measured in the Tertiary stock (Fig. 6), indicating that the dike has been reheated by the younger intrusive. Thermal demagnetization of BT-1 at temperatures of 400°C and higher causes scatter of the directions, indicating a PTRM acquisition temperature of less than 400°C.

Dike BT-2 has four whole-rock K-Ar ages from four samples taken along the length of the dike; they all yield a consistent age of about 730 m.y. (P. Mueller, 1972, personal commun.). The directions of magnetization for this dike from two sites (BT-2a, BT-2b) are stable to a.c. cleaning to 200 Oe and to thermal treatment to 400°C (Fig. 4). The virtual geomagnetic pole position for this dike is similar to those calculated for some early Paleozoic rocks (Black, 1964; Irving, 1964; and Larson and Mutschler, 1971), the 700-m.y.-old Franklin diabases of Canada (Fahrig and others, 1971) and the approximately 800-m.y.-old Frontenac Axis dikes (Park and Irving, 1972). Dike BT-2 is the only dike studied that contains well-grouped directions that are flat-lying to the west (typical of late Precambrian to early Paleozoic rocks); it is also the only one with a consistent late Precambrian age. We conclude, therefore, that this dike contains a *bona fide* late Precambrian direction of magnetization, and we accept the position of 7° N., 213° W. as a late Precambrian virtual geomagnetic pole (Table 1, Fig. 5).

Curie temperatures for the Beartooth dikes range from about 565° to 580°C, indicative of nearly pure magnetite (Fig. 7). This is supported by polished-section analysis (Appendix 1).

Bighorn Mountains, Wyoming

The Bighorn Mountains, similar tectonically to the Beartooth Mountains, bound the eastern margin of the Bighorn Basin. Mafic dikes in this region have been described by Osterwald (1959), Heimlich and Nelson (1972), and Ross and Heimlich (1972), and structural modifications of the Precambrian rocks are discussed by Osterwald (1959), Hoppin (1961, 1970), Hoppin and Palmquist (1965), Hudson (1969), Nelson (1969), and Starmer and Koucky (1970). Paleozoic sediments atop the Precam-

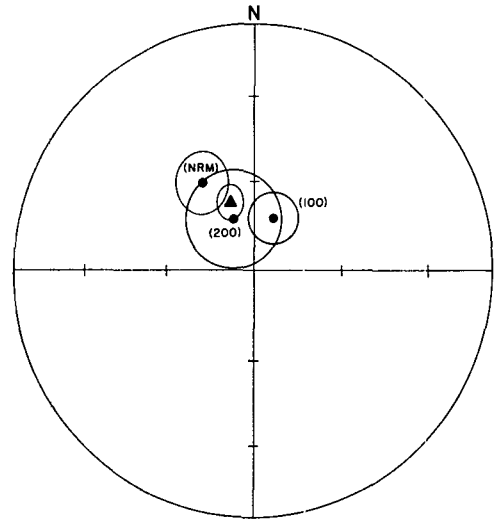


Figure 6. Mean directions of magnetization and 95 percent ovals of confidence of dike BT-1 (circles) and porphyry instrusive BT-9 (triangle). Peak fields of a.c. demagnetization in oersteds given in parentheses for BT-1. BT-9 after a.c. demagnetization in a 100 Oe peak field; equal-area projection; solid symbols denote downward.

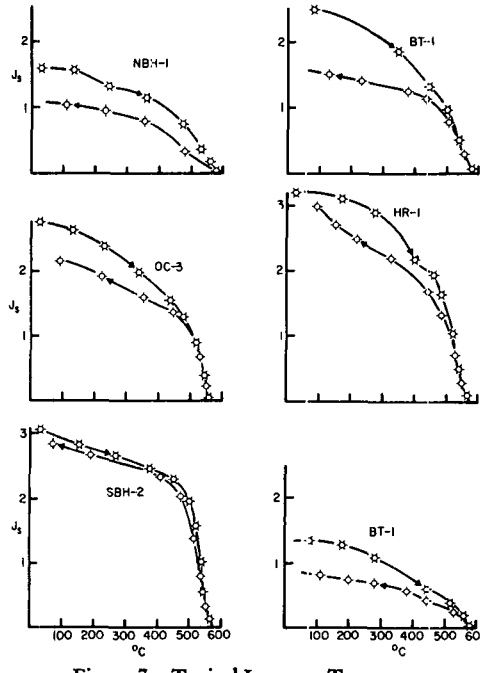


Figure 7. Typical Js versus T curves.

brian in this range are generally horizontal; the sampled mafic dikes are vertical or nearly so.

The crystalline basement of the Bighorn Mountains has been dated approximately as 3,000 m.y. old by K-Ar and Rb-Sr techniques (Heimlich and Banks, 1968). The whole-rock K-Ar ages on mafic dikes listed by Condie and others (1969) show a high degree of scatter with dates as old as 2,500 m.y.

The diabase dikes from the northern Bighorn Mountains (Fig. 8) appear to be unsuitable for paleomagnetic study. In general, the directions of magnetization of samples from these dikes show high dispersion, and results after a.c. demagnetization are commonly different than the results following thermal cleaning (dikes NBH-1, NBH-2, and NBH-6). However, in a few cases, there may be indications of magnetic directions reflecting the Precambrian geomagnetic field. The results from dikes NBH-1 (K-Ar age of 1,880 m.y.) and NBH-2 (K-Ar age of 2,020 m.y.), after thermal cleaning, and NBH-6 (K-Ar age of 1,590 m.y.) after a.c. demagnetization and thermal cleaning at 400°C give virtual geomagnetic pole positions centering at 67° S., 96° W., which are similar to those calculated for the 2,690 ± 93 m.y. Matachewan dikes (Strangway, 1964; Fahrig and others, 1965), the Stillwater Complex, between 2,450 ± 210 m.y. (Fenton and Faure, 1969) and 3,100 ± 200 m.y. (Bergh, 1970), and the unrecrystallized 2,550 m.y. Beartooth dike (after rotation). However, the high degree of scatter in the paleomagnetic results and the lack of adequate age control for the northern Bighorn dikes prevent their inclusion into a polar wandering path.

Three diabase dikes were sampled in the southern portion of the Bighorn Mountains. In general, the results from these dikes tended to show less dispersion than those from the northern portion of the Bighorn Mountains. The virtual paleomagnetic pole positions after a.c. and thermal demagnetization (near 52° S., 114° W.; Table 1) calculated for dike SBH-2 (K-Ar age of 1,390 m.y.) and SBH-3 (K-Ar age of 1,660 m.y.) are also similar to the paleomagnetic pole positions of the Matachewan dikes and the Stillwater Complex, and the virtual geomagnetic pole position of the 2,550-m.y. unrecrystallized Beartooth dike (BT-6). It is possible that the dikes in the Bighorn

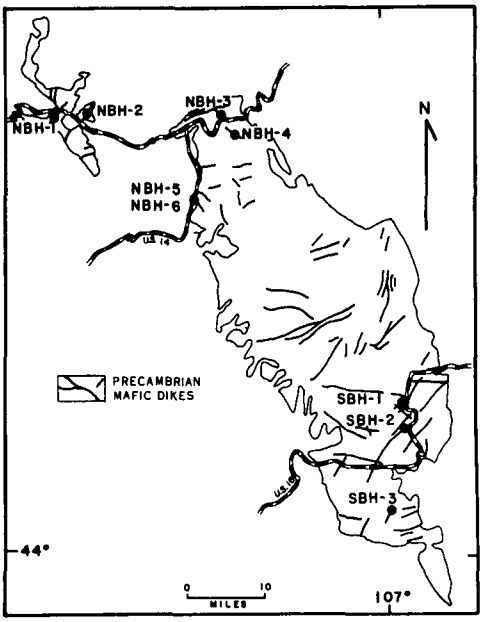


Figure 8. Location map of Northern and Southern Bighorn Mountains, Wyoming sampling sites (after Condie and others, 1969).

Mountains are older than 2,500 m.y. and paleomagnetically reflect the geomagnetic field of about 2,600 to 2,900 m.y. Because of the uncertainty in the reliability of the Bighorn Mountains paleomagnetic data, we have not used them in defining the oldest portion of our polar wandering path.

Curie temperatures for samples from both the northern and southern portions of the Bighorn Mountains range from about 550°C to 575°C, again indicative of nearly pure magnetite (Fig. 7). Polished-section analysis again supports this observation (Appendix 1).

Owl Creek Mountains, Wyoming

The Owl Creek Mountains which bound the southern portion of the Bighorn Basin basically consist of an uplifted block of Precambrian crystallines intruded by Precambrian mafic dikes. Nearby lower Paleozoic sediments to the east of the sampled area dip 5° to 12° NE., indicating that post-Precambrian deformation has been minimal. The planes of the dikes are vertical or nearly so, providing some evidence that Precambrian tectonism was also minor.

We sampled four dikes in the vicinity of Merritt Pass (Blondy Pass; Fig. 9), three of

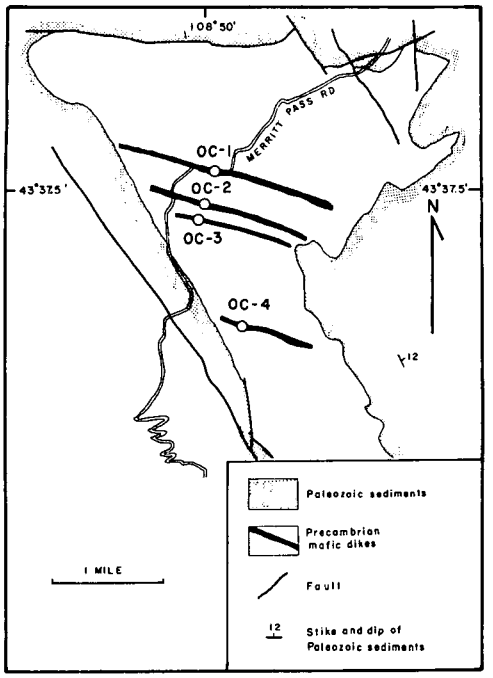


Figure 9. Location map of Owl Creek Mountains, Wyoming, sampling sites (after Masursky, 1950).

which survived the stability criteria. These three dikes yield consistent whole-rock K-Ar ages ranging from 1,910 to 2,090 m.y. (Condie and others, 1969).

Directions of magnetization of samples from dike OC-1 are extremely scattered and unacceptable. On the other hand, directions from dike OC-2 are reasonably well grouped and consistent at all demagnetization values (Fig. 4), yielding a virtual geomagnetic pole position at about 27° S., 111° W. A similar virtual pole (29° S., 120° W.,) was obtained from dike OC-3 by accepting directions of magnetization of two samples cleaned up to 400°C and three samples cleaned up to 200 Oe (Fig. 4). The high initial dispersion and variability in intensity observed in this dike are apparently caused by lightning-induced IRM which is difficult to remove equally at each demagnetization level. Consistent and very well grouped paleomagnetic directions were obtained from dike OC-4 (Fig. 4), yielding a virtual geomagnetic pole position of 23° S., 113° W. The mean paleomagnetic pole position from these dikes is located at 28° S., 112° W.; this pole is close to the rotated 2,160-m.y. pole from the recrystallized Beartooth dikes (Figs. 5 and 10).

As with the samples from the Beartooth and Bighorn Mountains, Curie temperatures from the Owl Creek dikes are about 550°C (see Fig. 7 and Appendix 1).

Grand Canyon, Arizona

The Precambrian rocks of the Grand Canyon, Arizona, region consist of the highly metamorphosed Vishnu Schist intruded by Zoroaster Granite (1,400 m.y. old) and overlain by the gently dipping Grand Canyon Series of quartzites, limestones, red shales, and basalt flows. The layered sequence has been intruded by diabase sills and dikes of the Rama Formation (Hendricks, 1972) which has been previously dated by K-Ar as 935 m.y. old (Ford and others, 1972) and 1,200 m.y. old (Cloud, 1971). We have obtained a whole-rock K-Ar age of 850 m.y. from a sill of the Rama Formation at Hance Rapids. It is possible that basic intrusives of more than one age are present in the Grand Canyon; but until more dates are available, we accept a 1,000-m.y. age for the Rama Formation.

We sampled four sills and a dike of the Rama Formation diabase intruding the Hakatai Shale and Bass Limestone in the vicinity of Hance

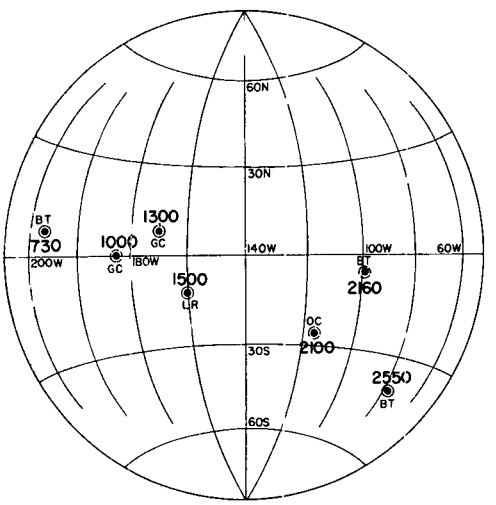


Figure 10. Virtual and paleomagnetic pole positions from our study; equal-area projection. Location abbreviations: BT, Beartooth Mountains; OC, Owl Creek Mountains; LR, Laramie Range; GC, Grand Canyon. Numbers refer to best estimate of age in millions of years. Number of site means involved in each pole position indicated by outer circle around each dot: complete circle, 1 site; three-segmented circle, 3 sites; 4-segmented circle, 4 sites; S, sediments.

Rapids in the bottom of the Grand Canyon (Fig. 11). Directions of remanence are tightly grouped and consistent at each demagnetization level within all sites, and the agreement between sites is remarkably good (Table 1; Fig. 4). The paleomagnetic pole position for the intrusives is located at 0°, 185° W. (Figs. 5 and 10). Applying a tilt correction to the mean directions of magnetization of the diabases based on the dip of the adjacent sediments caused the mean positions of the intrusives to scatter. From this observation, we conclude that intrusion of the Rama diabase in the Hance Creek area was an episode which took place either during the late stages of or after the tilting of the sediments of the Grand Canyon Series.

In addition, samples of unbaked Hakatai Shale were collected. The paleomagnetic pole calculated for the unbaked sediment, after applying a tilt correction, is located at 8° N., 170° W. (Table 1; Figs 5 and 11).

Curie temperatures range from 565° to 580°C for the diabase, indicative of essentially pure magnetite (Fig. 7). Polished-section data (Appendix 1) support this conclusion.

Anorthosite, Laramie Range, Wyoming

A large body of fresh anorthosite and syenite occupies the central portion of the Laramie Range northeast of Laramie, Wyoming (Fig. 12). It has been dated most recently by Hills and Armstrong (1971) by the K-Ar method at 1,500 m.y. We sampled the body at only one locality, a quarry about 30 mi northeast of Laramie. The directions are well grouped and consistent at every stage of demagnetization. The virtual geomagnetic pole position is 12° S., 160° W. (Table 1; Figs. 5 and 10).

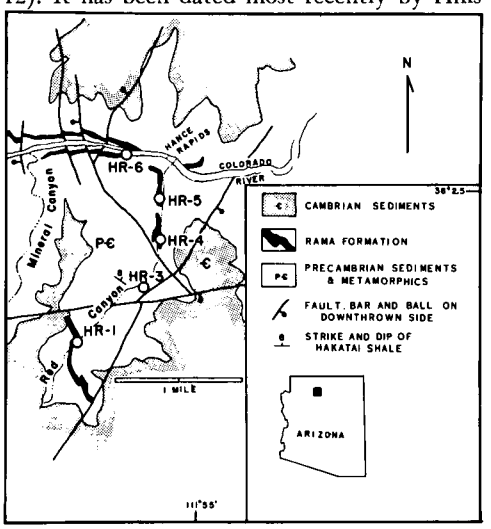


Figure 11. Location map of Hance Rapids, Grand Canyon National Park, Arizona, sampling sites (after Maxson, 1967).

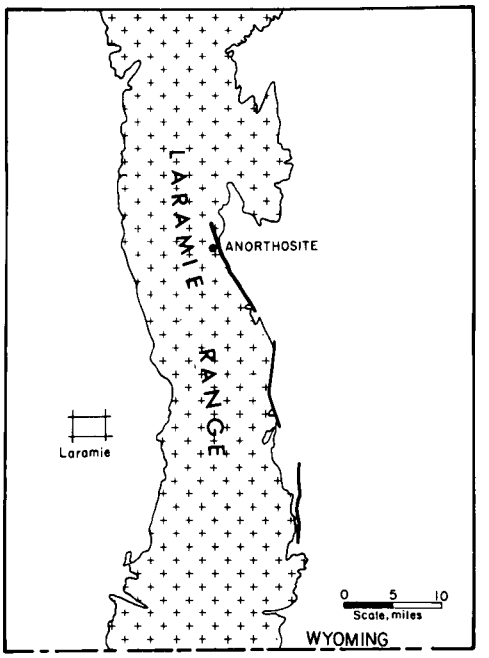


Figure 12. Location map of anorthosite collection site, Laramie Range, Wyoming.

Summary of Newly Obtained Results

Beartooth Mountains, Wyoming and Montana. Two dikes, which are dated at about 1,500 m.y. by both whole-rock K-Ar and Rb-Sr methods, yield a virtual geomagnetic pole at 4° S., 96° W. This position is so far from the well-accepted 1,500-m.y. direction as to require structural correction, and it forms the basis for structural rotations applied to all older dikes.

Using Rb-Sr dating by P. Mueller (1972, personal commun.) and K-Ar dating by Condie and others (1969), we conclude that the majority of Precambrian mafic dikes in the Beartooth Mountains belong to a set which was initially intruded about 2,550 m.y. ago and which were recrystallized and probably magnetically reset about 2,160 m.y. ago. A mean paleomagnetic pole for three dikes of the latter age is in the vicinity of 5° S., 49° W. If rotation

corrections are applied to these dikes, the pole moves to 5° S., 99° W. (Fig. 10).

A 2,550-m.y.-old virtual geomagnetic pole, based on one unrecrystallized dike, is located at 40° S., 27° W., before tilt correction and at 40° S., 77° W. after the application of the correction (Fig. 10).

A 730-m.y.-old Beartooth dike yields a virtual geomagnetic pole at 7° N., 213° W. (Fig. 10).

Bighorn Mountains, Wyoming. No reliable results were obtained from mafic dikes of the Bighorn Mountains. The data, however, are suggestive of paleomagnetic pole positions similar to those of the 2,500- to 3,000-m.y.-old Matachewan, Stillwater, and Beartooth intrusives.

Owl Creek Mountains, Wyoming. Three mafic dikes dated at about 2,000 m.y. yield a paleomagnetic pole position in the vicinity of 26° S., 115° W. (Fig. 10). This is quite close to the rotated 2,160-m.y. paleomagnetic pole from the Beartooth Mountains.

Grand Canyon, Arizona. Four diabase bodies with an estimated age of 1,000 m.y. in the Grand Canyon, Arizona, show a combined paleomagnetic position of 0°, 185° W. Unbaked sediment, into which the diabase is intruded, gives a paleomagnetic pole position of 8° N., 170° W. (Fig. 10).

Laramie Range, Wyoming. The virtual geomagnetic pole position for the 1,500-m.y.-old anorthosite near Laramie, Wyoming, is 12° S., 160° W. (Fig. 10).

NORTH AMERICAN PRECAMBRIAN APPARENT POLAR WANDERING PATH

There are now available more than 75 isotopically dated paleomagnetic positions for the North American Precambrian. Most occur in the range of 1,400 to 1,000 m.y. When pole positions covering a restricted time span are plotted, they commonly show a great deal of scatter. This lack of grouping is to be expected for Precambrian rocks for the following reasons.

Accurate dates of rocks as old as 3,000 m.y. by isotopic methods (particularly K-Ar) are essentially unobtainable because of possible hydrothermal and weathering alteration, subtle or multiple metamorphic events, and (or) reheating. K-Ar dating of single rock samples that appear to be of the same dike set commonly yields ages which are scattered and which appear to be young when compared with Rb-Sr isochron dates for the same rocks. An example is provided by comparison of the ages of dikes in the Beartooth Mountains. K-Ar ages (Condie and others, 1969) show scatter from 1,010 to 2,160 m.y. for rocks dated by Rb-Sr methods at 2,550 m.y. (P. Mueller, 1972, personal commun.).

One prime use of K-Ar ages for Precambrian rocks, particularly on single samples of whole rock, is to show whether the rocks have been greatly affected by some thermal event in the recent past. The K-Ar age of 1,010 m.y. for dike BT-1 (Rb-Sr age of 2,550 m.y.) displays this phenomenon. There is no doubt that reheating of this dike by the nearby intrusion of a Tertiary stock has partially reset the radioisotope clock and fully reset the paleomagnetic directions.

K-Ar ages can be used in defining a later period of recrystallization. Three of the Beartooth dikes showed similar recrystallization textures and gave essentially the same paleomagnetic directions. K-Ar dates of the three are 1,690, 1,960, and 2,160 m.y.; whereas Rb-Sr whole-rock isochrons indicate an age of 2,550 m.y. The maximum age of 2,160 m.y. probably approximates the time of recrystallization.

In addition, structural deformation of the Precambrian rocks used in constructing a polar wandering path has almost certainly occurred but is often subtle and difficult to reconstruct. Verticality of dikes provides only a crude measure of movement. In the Beartooth Mountains, we have a reasonable but tentative indication of rotational deformation provided by the displaced 1,500-m.y. directions of dikes BT-7 and BT-8. The exact amount of rotation cannot be accurately known, however, inasmuch as the amount of geomagnetic secular dispersion is unknown.

Another reason for the scatter of pole positions involves secondary magnetization, most commonly a chemical remanent magnetization (CRM). The processes and manifestations of chemical remagnetization are very poorly understood at present; in some instances, the original remanence of certain remagnetized rocks may not be recognizable. To aid in the determination of remagnetization, both a.c. and thermal demagnetization for all samples are essential. Few of the paleomagnetic poles for the North American Precambrian are based on samples which have undergone these two treatments. Even when both treatments have been

applied, there is no certainty that the effects of secondary remagnetization can be determined or removed. In our investigation of Precambrian rocks in the Front Range, Colorado, we have commonly encountered granitic rocks which have been chemically remagnetized during the last 70 m.y.; yet the overprint cannot be removed by either a.c. or thermal cleaning. In an investigation of Precambrian rocks in the Wind River Range, we have sampled dikes which have been thoroughly remagnetized by development of hematite. Only in those cases where the remagnetization is in its early stages, can it be determined and separated from initial components by a.c., thermal, and possibly chemical cleaning. Hematization is commonly a subtle process which is very difficult to definitely identify microscopically until the hematite is well developed.

In view of these uncertainties, we feel that a realistic approach to constructing a Precambrian polar wandering path for North America requires statistical treatment of the data, rather than drawing lines through individual poles or groups of poles. The data used in the analysis must be reliable; that is, acceptable criteria must be applied both as to the reliability of the age of the rocks and their paleomagnetism. To reduce bias, multiple data points from rocks of the same type, of the same age, and in a relatively restricted geographic region should be averaged. We have analyzed existing data in this manner and present our results in Figure 13. Because of the paucity of data points for some age spans, and because the isotopic ages are known (except rarely) to ± 0.1 b.y. at best, we have used a modified running mean method, choosing 0.3 b.y. to 0.5 b.y. as sufficient spans for mean calculations. The spacing of the limits becomes less regular in the analysis of the older paleomagnetic pole positions because of the sparseness of the data. We have calculated mean paleopoles and 95 percent confidence cones for times 0.7 to 1.0, 1.0 to 1.2, 1.2 to 1.4, 1.4 to 1.8, 1.8 to 2.2, 2.2 to 2.6, and 2.6 to 2.9 b.y. (Fig. 13). Analysis of Precambrian paleomagnetic data for North America in this manner reveals a simple S-shaped belt of apparent polar wandering. The belt is bounded by smoothed lines approximately tangent to the 95 percent confidence cones. One striking feature of the path is the obvious absence of the large-scale loops which have characterized previous interpretations. Furthermore, our belt demonstrates a relatively uniform rate of Precambrian polar

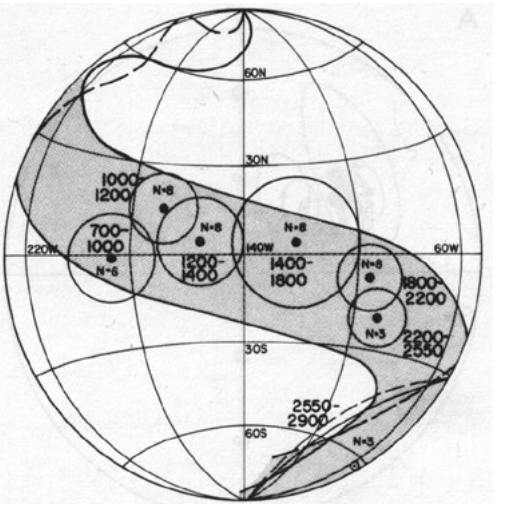


Figure 13. Mean paleomagnetic pole positions and 95 percent confidence circles computed, in the age ranges given in millions of years, for the pole positions shown in Figure 5; N = number of samples in each age range, equal-area projection. Stippled pattern indicates generalized apparent polar wandering path for Precambrian time.

wander (about 0.5° per 5 m.y.), also unlike previous interpretations.

It might be argued that if loops did truly exist, our technique would tend to minimize their existence. This is true at the present but will be less so as more well-dated, well-cleaned directions become available.

It is interesting to note that all loops recently postulated (Spall, 1971a; Robertson and Fahrig, 1971; Irving and others, 1972c) basically represent rapid excursions from and subsequent return to our belt of polar wander. If apparent polar wandering during short episodes represents continental motions and not manifestations of a nondipole field, it is curious that the land mass should swing rapidly away, make a large latitudinal diversion, and then return nearly to the same orientation as before. This behavior could also result from acceptance of several deviant data points so that large swings of the polar path were required for their inclusion. It is appropriate, therefore, to re-examine the evidence for the wander loops.

The Keweenawan trend (1,100 to 1,000 m.y.) of DuBois (1962) and Spall (1971a) is based largely on data from Keweenawan rocks of the Lake Superior region. Much of the data on which DuBois based his loop was from samples

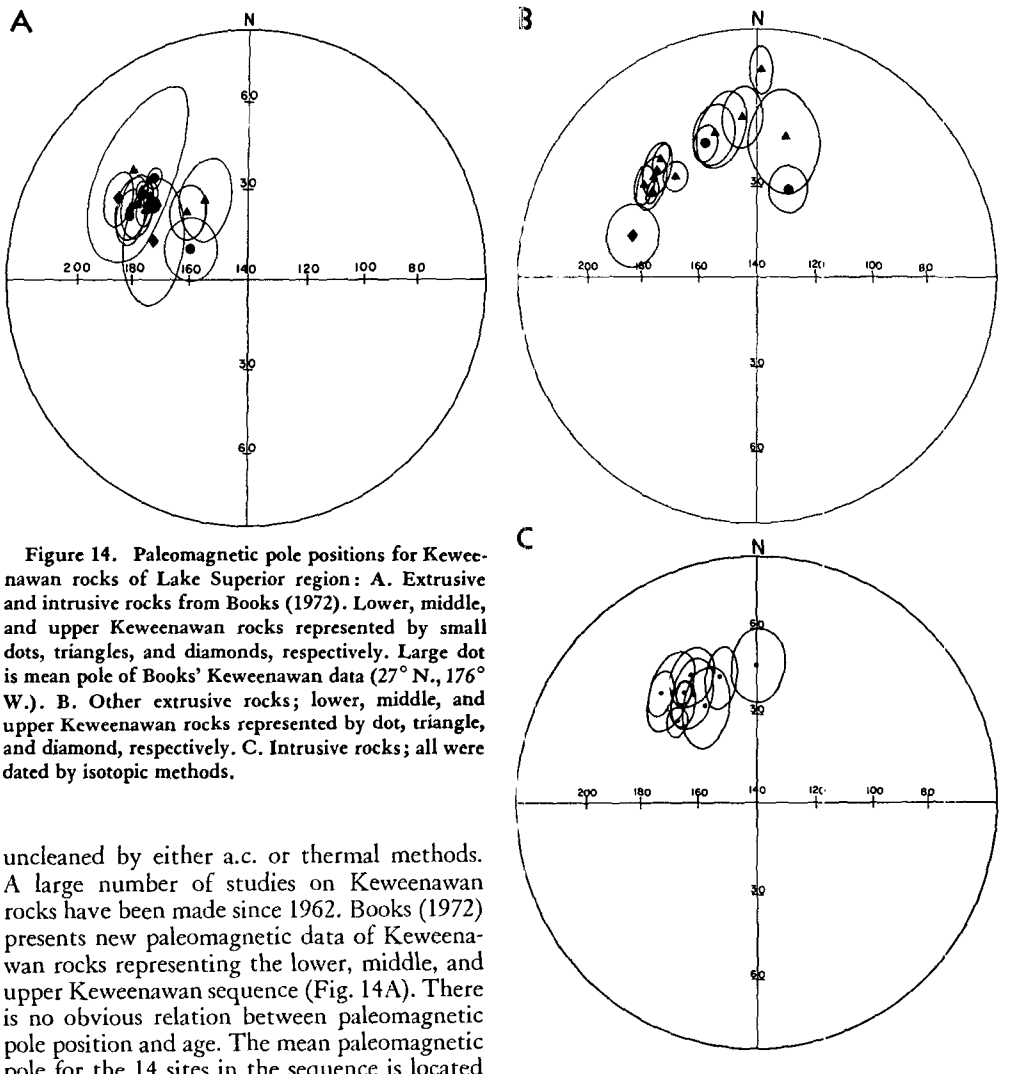


Figure 14. Paleomagnetic pole positions for Keweenawan rocks of Lake Superior region: A. Extrusive and intrusive rocks from Books (1972). Lower, middle, and upper Keweenawan rocks represented by small dots, triangles, and diamonds, respectively. Large dot is mean pole of Books' Keweenawan data (27° N., 176° W.). B. Other extrusive rocks; lower, middle, and upper Keweenawan rocks represented by dot, triangle, and diamond, respectively. C. Intrusive rocks; all were dated by isotopic methods.

uncleaned by either a.c. or thermal methods. A large number of studies on Keweenawan rocks have been made since 1962. Books (1972) presents new paleomagnetic data of Keweenawan rocks representing the lower, middle, and upper Keweenawan sequence (Fig. 14A). There is no obvious relation between paleomagnetic pole position and age. The mean paleomagnetic pole for the 14 sites in the sequence is located at 23.6° N., 176.2° W. ($\delta p = 4.0°$, $\delta_m = 7.0°$), which is within the limits of our polar wandering belt. A compilation of paleomagnetic pole positions of extrusive Keweenawan rocks obtained prior to Books' study is shown in Figure 14B. Although dispersion is greater among these positions, again there is no unambiguous relation of pole location and age. If one considers the Keweenawan intrusives of the Lake Superior region (which yield ages around 1.0 to 1.1 b.y.), no clear-cut streaking is discernible either (Fig. 14C). Rather, there is a fairly tight clustering of poles in the vicinity of 30° N., 167° W., close to the mean direction of the Keweenawan flows. These results are consistent with the conclusions of Silver and Green (1972) who have extended their U-Pb isotope-dating studies to a large suite of Keweenawan rocks including flows and intrusives. They conclude that all Keweenawan activity represents a short-lived pulse of volcanic activity, 20 ± 10 m.y. long, that continued from 1,120 to 1,140 m.y. ago. Maximum Rb-Sr ages that have been determined for these rocks correspond to the 1,120- to 1140-m.y. age.

Robertson and Fahrig (1971) use paleomagnetic poles from the Keweenawan intrusives

to better define the loop which they term the Great Logan Loop (Fig. 1). Although cross-cutting relations do provide some measure of relative age classification of the various intrusives, they have to rely on radiometric dating to indicate the close timing necessary for the Logan Loop path. They place confidence in differences of whole-rock K-Ar ages of 30 m.y. (3 percent of age) and 50 m.y. (5 percent of age) in distinguishing between the different pole positions. In addition, their assigned age for the Logan Sills, 1,050 m.y., is based on a spread of K-Ar ages from 730 ± 50 m.y. to 1,000 ± 140 m.y.; their assigned age for the Normal dikes, 1,020 m.y., is based on a spread of K-Ar ages of 993 ± 35 m.y. to 1,055 ± 40 m.y. The Aillik dikes (Fahrig and Larochelle, 1972) have been included in the Logan Loop because of their paleomagnetic pole position (27° N., 136° W.), a position in accord with an age of 1,150 m.y. However, the K-Ar ages of the Aillik dikes range from 685 ± 32 to 995 ± 13 m.y.

To reduce bias resulting from consideration of similar rock types of similar ages from the same region, we have chosen in our analysis two mean paleomagnetic pole positions, one from the flows and one from the dikes, as representative of the Lake Superior basic rocks of 1,100 to 1,000 m.y. By doing so, the density of paleopoles of 1,200 to 1,000 m.y. is greater within the newly proposed wander path than outside it (Fig. 13). Hence we present an alternative interpretation to the 1,200- to 1,000-m.y. wander loop: the polar wandering during that time was simple and of relatively uniform rate. We feel that if a loop indeed exists, proof of its existence should be obtained from outside of the Lake Superior region.

With reference to the 2,500- to 1,500-m.y. Animikie-Sudbury trend of Spall (1971a), only the Wind River dikes (Spall, 1971b) have been treated by a.c. and thermal cleaning; uncertainty exists on the interpretation of structural correction to be used in defining the Sudbury Irruptive paleomagnetic poles (Hood, 1961; Sopher, 1963; and Larochelle, 1969), and the results from the Marathon dikes are based on nine samples, treated only by single-axis a.c. demagnetization techniques and dated by a single whole-rock determination (Fahrig and others, 1965). Paleomagnetic data are much more abundant on either side of the Animikie-Sudbury loop (Spall, 1971a).

The path of Irving and others (1972c) in the time span of 2,500 to 1,500 m.y. lacks the northward swing of Spall. Instead, it loops far to the east to include the paleomagnetic-pole position from the Et-Then Group. These data appear to be an insufficient basis for delineation of a loop, especially since the Et-Then paleomagnetic pole position is entirely from red sediments.

PRECAMBRIAN APPARENT POLAR WANDERING AND CONTINENTAL DRIFT

It is well established that the forces as described in plate tectonic theory have been operating since the beginning of the Mesozoic. A crucial question facing geologists and geophysicists today is whether identical processes were operative in the pre-Mesozoic. Paleomagnetism will play a major role in attempting to resolve this question (Briden, 1967; McElhinny and Briden, 1971; McElhinny, 1967; Creer, 1972; Irving and others, 1972a; Larson and others, 1972; and Spall, 1972).

Spall (1972) has suggested that North America and southern Africa experienced differential movement throughout much of the Precambrian. However, the data from southern Africa is so scanty that it appears premature at this time to try to derive an apparent polar wandering path for that region.

CONCLUSIONS

Reliable virtual and paleomagnetic pole positions have been identified from Precambrian rocks (particularly mafic dikes) as old as 2,600 m.y. within Wyoming, Montana, and Arizona. When these data are added to the compilation of other well-dated poles from igneous rocks of North America, a belt of poles is indicated which is simple in form and of nearly uniform wander velocity. The existence of large-scale wander loops is unsupported, or at least, questionable.

ACKNOWLEDGMENTS

We thank P. Mueller for helpful discussions on the age relations of the Beartooth dikes, L. LaFountain for assistance in the field, and M. Mulhern for helping with the thermomagnetic analysis. The National Park Service kindly gave permission to collect in the Grand

Canyon. This work was supported by National Science Foundation Grant GA-27663.

APPENDIX 1. BRIEF SUMMARY OF POLISHED-SECTION ANALYSIS (PALEOMAGNETICALLY IMPORTANT MINERALS ONLY)

Beartooth Mountains

Dike BT-1 has apparently lost its original magnetite due to alteration. Dikes BT-2, BT-6, and BT-7 are characterized by large (as much as 900 μ) magnetite phenocrysts cut by ilmenite lamellae. The magnetite has varying degrees of the same mottled texture seen in the Hance Rapids specimens. Dikes BT-3 and BT-4 are distinguished by small (1 to 10 μ) magnetite phenocrysts in iron silicates or as intergranular fillings. Dike BT-5 possesses very abundant, very small (1 μ) magnetite phenocrysts in iron silicates and large magnetite phenocrysts (>400 μ) laced by ilmenite exsolution lamellae. Dikes BT-1, BT-3, BT-4, and BT-5 show signs of recrystallization. Curie temperatures range from about 565° to 580°C, indicative of nearly pure magnetite.

Bighorn Mountains

Northern Bighorns. In general, titanomagnetites which underwent high-temperature oxidation were originally present as evidenced by the presence of ilmenite lamellae alternating with nonopaques or hematite. The primary magnetite is now replaced except for small (1 to 10 μ) grains present in iron silicates.

Southern Bighorns. The Southern Bighorns are similar to the Northern Bighorns, although more primary magnetite is present.

Owl Creek Mountains. Dikes OC-2, OC-3, and OC-4 possess large (>250 μ) magnetite grains containing ilmenite exsolution lamellae and abundant small (1 to 10 μ) grains disseminated through silicates. This magnetite is homogeneous and fresh with little sign of alteration. In contrast, all magnetite in OC-1 has been replaced by leucoxene and shows signs of massive alteration.

Grand Canyon, Arizona. Intrusives HR-1, HR-4, HR-5, and HR-6 are characterized by large (>300 μ) magnetite grains with ilmenite lamellae. The magnetite has a mottled texture which varies from very subtle to very pronounced. We suggest this mottling is due to relatively low-temperature oxidation (in contrast to the temperatures of formation of the usual ilmenite lamellae) to ilmenite and magnetite and (or) hematite. Numerous small magnetite grains (1 to 10 μ) are present disseminated in iron silicates. Hematite is common as a silicate alteration.

HR-3 is fine-grained hematitic sediment.

APPENDIX 2. SAMPLE LOCALITY DESCRIPTIONS

Beartooth Mountains

BT-1. Highway 212 roadcut exposure of quartz dolerite dike (Prinz, 1964) which trends N. 25° W. Mt. Maurice, Montana-Wyoming, 15′ quadrangle.

BT-2a. Coarse-grained, spheroidally weathered dolerite dike 6 m thick. Collected about 20 m west of Highway 212 and 0.3 km northwest of summit bench mark 10,936 ft. Deep Lake, Wyoming, 15′ quadrangle.

BT-2b. Natural exposure of dolerite dike near south shore of Tiel Lake less than 1 km north of the Montana-Wyoming boundary. Alpine, Montana-Wyoming, 15′ quadrangle.

BT-3. Roadcut exposure of 2-m-thick dike on east side of Highway 212, about 0.2 km northwest of bench mark 10,049 ft. Archean metadolerite of Prinz (1964). Deep Lake, Wyoming, 15′ quadrangle.

BT-4. Dike, about 15 m thick, collected at three localities: in quarry 50 m east of Highway 212, 0.3 km northeast of elevation mark 9,841; near the outlet of lake about 3.0 km northwest of quarry; and west of Sheepherder Lake about 4.0 km, northwest of quarry. Archean metadolerite of Prinz (1964). Deep Lake, Wyoming, and Beartooth Butte, Wyoming, 15′ quadrangles.

BT-5. Natural exposure of mafic dike at southern end of Long Lake adjacent to bench mark 9,655 ft. Archean metadolerite of Prinz (1964). Beartooth Butte, Wyoming, 15′ quadrangle.

BT-6. Roadcut exposure of thick (60 m) coarse-grained dike. Dolerite dike of Prinz (1964); Beartooth Butte, Wyoming, 15′ quadrangle.

BT-7. Natural exposure of mafic dike about 0.4 km due east of the head of Beartooth Creek. Beartooth Butte, Wyoming, 15′ quadrangle.

BT-8. Natural exposure of mafic dike about 1 km north of Tiel Lake. Alpine, Montana-Wyoming, 15′ quadrangle.

BT-9. Samples drilled in place from roadcut exposure of porphyritic Tertiary intrusive about 0.8 km north along Highway 212 from bench mark 9,528 ft. Mount Maurice, Montana-Wyoming, 15′ quadrangle.

Northern Bighorn Mountains

NBH-1. Natural exposure of a 30-m-thick mafic dike 60 m west of Bald Mountain campground. Bald Mountain, Wyoming, 7½′ quadrangle.

NBH-2. Natural exposure of a 30-m-thick mafic dike 4.5 km along Highway 14, east of Bald Mountain campground turnoff and 40 m north of the highway. Bald Mountain, Wyoming, 7½′ quadrangle.

NBH-3. Prospect pit exposure of a 10-m-thick dike about 10 km east on Highway 14 from Sibley

Lake and 70 m northwest of the highway opposite bench mark 7,982. Skull Ridge, Wyoming, 7½′ quadrangle.

NBH-4. Roadcut exposure about 2 km south along dirt road which leaves Highway 14 at bench mark 7,982. Skull Ridge, Wyoming, 7½′ quadrangle.

NBH-5. Thin (1 m), fine-grained mafic dike exposed in west-side roadcut in Granite Pass, Highway 14. Granite Pass, Wyoming 7½′ quadrangle.

NBH-6. Mafic dike, 25 m thick, poorly exposed in bulldozer scraping adjacent to roadcut, 30 m north of NBH-5. Granite Pass, Wyoming, 7½′ quadrangle.

Southern Bighorn Mountains

SBH-1. Mafic dike, 10 m thick, exposed in dirt road to Lynx Park off Highway 16. Hunter Mesa, Wyoming, 7½′ quadrangle.

SBH-2. Natural exposure of a ridge-forming mafic dike, 35 m thick, 100 m south of dirt road to E-La-Ka-Wee Camp off Highway 16. Caribou Creek, Wyoming, 7½′ quadrangle.

SBH-3. Natural exposure of 50-m-thick mafic dike about 10 m east of Hazelton Road, 14 km south of Hazelton Road–Highway 16 junction. Sheridan, Wyoming–Montana 2° sheet.

Owl Creek Mountains

OC-1. Roadcut exposure of fine-grained dike about 10 m thick trending N. 80° W. Anchor Reservior, Wyoming, 7½′ quadrangle.

OC-2. Natural exposure of fresh, 30-m-thick mafic dike collected 0.6 km southeast from OC-1. Bargee, Wyoming, 7½′ quadrangle.

OC-3. Natural exposure of fresh mafic dike collected about .27 km south-southeast from OC-2. Bargee, Wyoming, 7½′ quadrangle.

OC-4. Stream canyon exposure of fresh, fine- to medium-grained mafic dike about 10 m thick, about 0.7 km southeast of Merritt Pass, Bargee, Wyoming, 7½′ quadrangle.

Laramie Range Wyoming

IM-1. Natural exposures collected at Iron Mountain, 47 mi northeast of Laramie by road in sections 22, 23, and 27 of T. 19 N., R. 71 W., Albany County. Goat Mountain, Wyoming, 7½′ quadrangle.

Grand Canyon Arizona

HR-1. Diabase sill and low-angle dike which cuts the Hakatai Shale near its base, about 2.5 km south of Hance Rapids in Red Canyon.

HR-3. Unbaked Hakatai Shale, about 60 m stratigraphically below base of the Shinumo Quartzite. Attitude: N. 55° W. 8° NE. Exposure in Red Canyon about 1.5 km south of Hance Rapids.

HR-4. Samples of diabase and baked Hakatai Shale from a sill-dike near the mouth of Red Canyon about 0.5 km south of Hance Rapids. Sill is about 1 m thick.

HR-5. Sill (3 m thick) in ravine 0.2 km south of Hance Rapids in Red Canyon. Samples of diabase and baked shale.

HR-6. Course-grained diabase sill (70 to 100 m thick) intruding Bass Limestone about 1 km west of Hance Rapids; adjacent to Colorado River channel on south side. Whole-rock K-Ar age 0.85 b.y.

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